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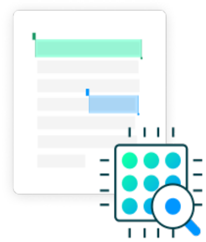

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CHAPTER 1. INTRODUCTION

1.1 Background of the Study

Segmentation of medical images, specifically brain tumor segmentation based on Magnetic Resonance Imaging (MRI) scans, is at the core of contemporary clinical oncology (S. Cai et al., 2022; X. Liu et al., 2021). Segmentation is valuable because it conveys useful information for diagnosis, treatment planning, and monitoring disease extent by accurately defining tumor outline and characteristics (X. Liu et al., 2021; Tang, 2019). Segmenting an image into clearly demarcated regions according to standard features is necessary to decomplicate complex medical images so they can be more easily interpreted. It enhances our interpretation of these images and minimizes the opportunities for errors in diagnosis (Handels et al., 2013; X. Liu et al., 2021). Over the past few years, CAD has advanced further than imagined with the evolution of deep learning (specifically via Convolutional Neural Networks - CNNs). These improvements help create segmentation of medical images that are more accurate, consistent, and trustworthy than methods with traditional image processing (Litjens et al., 2017). Well-known network architectures like U-Net (Ronneberger et al., 2015) have led these advancements. Further, its derivatives are now standard organ and lesion segmentation procedures in various imaging modalities, including MRI for tumor analysis of brain tumors (S. Cai et al., 2022; Moorthy & Gandhi, 2022; Tang, 2019). The improvements achieved in the deep learning models have tremendous promise to improve the speed and uniformity of tumor segmentation. This automation would significantly reduce the workload for radiologists and other medical professionals, enabling them to concentrate more on patient care (Moorthy & Gandhi, 2022; Tang, 2019).

1.2 Statement of the Problem

Despite these strides, challenges remain. A recent advancement in image segmentation is the Attention Synergy Network (AS-Net), proposed by K. Hu et al. (2022) for skin lesion segmentation. AS-Net effectively utilizes spatial and channel attention mechanisms to improve segmentation performance. Nonetheless, the initial study recognized limitations regarding computational requirements and possible generalizability concerns due to its use of the comparatively resource-intensive VGG16 encoder architecture. This computational inefficiency is a limitation to real-world deployment in clinical environments where resources are limited and quick processing is preferable.

Therefore, this research addresses the central problem of the original AS-Net architecture's suboptimal computational efficiency when considering its potential application to MRI brain tumor segmentation. While the attention mechanisms show promise, the VGG16 encoder limits their practicality. This study investigates whether adapting AS-Net with more modern, lightweight encoder architectures can maintain or improve segmentation accuracy while significantly enhancing computational efficiency.

This study seeks to answer the following research questions:

1. How can lightweight encoder architectures, specifically MobileNetV3 and EfficientNetV2, be effectively integrated into the existing AS-Net framework for MRI brain tumor segmentation?

2. How do the AS-Net variants using MobileNetV3 (Large and Small) and EfficientNetV2 (B0 and B1) encoders compare to the original AS-Net with a VGG16 encoder in terms of:

    - Segmentation accuracy (measured by Dice Coefficient/F1 Score, IoU/Jaccard Index, Precision, Recall, Binary Accuracy)?
    - Computational efficiency (measured by total training time)?

3. Does the synergy of spatial and channel attention mechanisms within AS-Net remain effective for segmenting brain tumors in MRI data when paired with different encoder backbones?

1.3 Objectives of the Study

The primary purpose of this research is to adapt and optimize the Attention Synergy Network (AS-Net) architecture for MRI-based brain tumor segmentation by systematically comparing the performance and efficiency of different encoder backbones.

Specifically, this study aims to:

- Assess the incorporation of light encoder architectures (MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, EfficientNetV2-B1) into the AS-Net framework as substitutes for the base VGG16 encoder.
- Quantify and compare segmentation performance (in terms of metrics such as Dice Coefficient, IoU, Precision, Recall) and computational efficiency (train time) of the AS-Net variants with these various encoders.
- Assess the effectiveness of the AS-Net's spatial and channel attention mechanisms when applied to the specific MRI brain tumor segmentation domain.
- Provide evidence-based recommendations regarding the most suitable encoder architecture for balancing accuracy and efficiency within the AS-Net framework for this application.

1.4 Significance of the Study

This research holds significant potential for advancing knowledge in medical image analysis, particularly concerning deep learning applications for brain tumor CAD. Adapting and optimizing the AS-Net architecture (K. Hu et al., 2022) for MRI brain tumor segmentation contributes valuable insights into designing and applying attention-based deep learning models for complex medical imaging tasks.

The exploration of lightweight encoders like MobileNetV3 and EfficientNetV2 within the AS-Net framework addresses a critical need for practical, deployable AI solutions in healthcare. By demonstrating the feasibility of achieving high segmentation accuracy with reduced computational cost, this study can facilitate the adoption of advanced segmentation tools in real-world clinical workflows, potentially operating on resource-constrained hardware or enabling faster processing times.

The results are likely to assist various stakeholders:

1. **Radiologists and Clinicians:** Accessing more efficient, precise automated segmentation solutions can enhance confidence in diagnosis, make treatment planning easier, and enable more accurate monitoring of tumor response. It all cumulatively enables improved patient care and management.

2. **Patients:** More accurate and efficient segmentation can lead to earlier detections, more precise treatments, and potentially better health outcomes.

3. **Researchers and Developers:** This work provides empirical evidence on the performance trade-offs of different encoders within an attention-based framework, informing future development of deep learning models for medical imaging. Adapting the attention synergy mechanism offers insights into the role of attention in extracting relevant features from MRI data.

Ultimately, this research aims to bridge the gap between cutting-edge deep learning techniques and practical clinical applications, contributing to developing more accessible and practical tools for brain tumor analysis.

1.5 Scope and Limitation of the Study

1.5.1 Scope

This research is specifically focused on:

- Adapting and evaluating the Attention Synergy Network (AS-Net) architecture.

- The task of binary brain tumor segmentation (tumor vs. background) using MRI data. Multi-class segmentation (differentiating tumor sub-regions) is outside the scope.
- Comparing the performance of five specific encoder backbones: VGG16, MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, and EfficientNetV2-B1. Other potential encoders are not evaluated. Input sizes were 224x224 for VGG16, MobileNetV3-Large/Small, and EfficientNetV2-B0, and 240x240 for EfficientNetV2-B1.
- Utilizing the BraTS 2023-GLI (Task 1) dataset. The original NIfTI files were preprocessed into 2D HDF5 slices for efficient loading. The study focuses only on the glioma segmentation task (Task 1) from the BraTS 2023 challenge and does not involve acquiring raw MRI data.
- Evaluating performance based on standard segmentation metrics (Dice, IoU, etc.) and computational efficiency based on training time on specific hardware.

1.5.2 Limitations

The study acknowledges the following limitations:

- **Dataset Specificity:** Findings are based on the BraTS 2023-GLI (Task 1) dataset. The generalizability of the results to other datasets (including other BraTS 2023 tasks) with different patient populations, tumor types, or imaging protocols requires further investigation.
- **Data Characteristics:** The quality, resolution, and potential artifacts within the original NIfTI files, as well as the specific method used for preprocessing into HDF5 slices, could influence model performance.
- **Computational Resources:** Training and evaluation were conducted on specific hardware (NVIDIA Tesla V100). Training times are relative and would dif-

fer on other systems. The scale of experimentation (e.g., number of runs, extent of hyperparameter tuning) was constrained by available resources.

- **Hyperparameter Selection:** For comparability, the study used a fixed set of primary hyperparameters (e.g., learning rate, loss weights) across models. Individual tuning for each encoder might yield slightly different performance rankings.
- **Implementation Details:** Specific choices regarding fine-tuning strategy (which layers to freeze/unfreeze) and skip connection points could impact results.

CHAPTER 2. REVIEW OF RELATED LITERATURE

This chapter reviews the literature relevant to MRI brain tumor segmentation using deep learning, providing the context for the current study. It begins by discussing the importance and challenges of brain tumor segmentation using MRI. It then covers the evolution from traditional methods to deep learning approaches, focusing on CNNs and the U-Net architecture. Subsequently, it delves into the concept and application of attention mechanisms within deep learning for medical image segmentation. The chapter then examines explicitly the Attention Synergy Network (AS-Net), detailing its architecture and identifying its limitations related to computational efficiency. Finally, it explores the significance and characteristics of lightweight encoder models, such as MobileNetV3 and EfficientNetV2, as possible solutions to shatter the efficiency bottleneck of AS-Net for practical application in brain tumor segmentation.

2.1 MRI Brain Tumor Segmentation: Importance and Challenges

Magnetic Resonance Imaging (MRI) is the first line in diagnosing and assessing brain tumors due to its excellent soft-tissue contrast and the potential for acquiring multi-parametric information with different sequences (e.g., T1, T1-contrast enhanced (T1ce), T2, FLAIR) (Gordillo et al., 2013). Accurate segmentation of brain tumors from MRI volumes is of clinical importance for numerous reasons: it is useful in diagnosis and grading, useful in surgical planning and radiotherapy treatment planning, useful in quantitative monitoring of response to therapy, and useful in volumetric analysis for follow-up of disease progression (Bakas et al., 2021; W.-L. Cai & Hong, 2018; Tang, 2019). Manual segmentation is a time-intensive, labor-intensive process that requires much expertise and is fraught with inter- and intra-rater variability (Bakas et al., 2021). This has necessitated the creation of automated and semi-automated segmentation methods.

Automatic brain tumor segmentation remains challenging despite much research for a variety of reasons (Gordillo et al., 2013; Litjens et al., 2017):

- **Tumor Heterogeneity:** Gliomas, the target of the BraTS challenge, and other brain tumors are extremely heterogeneous in size, shape, location, and visual appearance across subjects and even within one tumor (e.g., edema, necrotic core, enhancing tumor) (Bakas et al., 2021; Ranjbarzadeh et al., 2021).
- **Indefinable Boundaries:** Tumor borders can be infiltrative and indistinct, and therefore hard to define precisely.
- **Class Imbalance:** Tumor regions usually occupy a small fraction of the total brain volume, leading to highly imbalanced classes, which can make normal training procedures cumbersome (Havaei et al., 2017; Isensee et al., 2018).
- **Data Scarcity and Annotation Cost:** Acquiring large-scale, multi-modal MRI datasets with accurate, voxel-level ground truth annotations is expensive and time-consuming, limiting the data available for training supervised deep learning models (Ullah et al., 2023).
- **Computational Demands:** It takes significant computational resources (GPU memory, computational power) to process large 3D MRI volumes, particularly with sophisticated deep learning models (Havaei et al., 2017).

2.2 Evolution of Segmentation Methods

2.2.1 Traditional Approaches

Early attempts at automated brain tumor segmentation relied on traditional image processing techniques. These included intensity-based methods like thresholding (Gordillo et al., 2013), region-based methods like region growing (Melouah & Layachi, 2018), edge-based methods using active contours or level sets (Caselles et al., 1997; Pierre et al., 2021), and methods based on statistical models or machine learning classifiers using hand-crafted features (Gordillo et al., 2013). While useful, these methods often struggled with the heterogeneity of tumors, required significant parameter tuning, and were sensitive to noise and imaging artifacts (W.-L. Cai & Hong, 2018).

2.2.2 Deep Learning and Convolutional Neural Networks (CNNs)

The emergence of deep learning, especially CNNs, has significantly improved the state-of-the-art in medical image segmentation (Litjens et al., 2017; X. Liu et al., 2021; Moorthy & Gandhi, 2022). CNNs learn hierarchical features automatically from image data without the need for feature engineering by hand and capture intricate spatial patterns useful for segmentation.

The U-Net architecture (Ronneberger et al., 2015) has especially impacted biomedical image segmentation. Its skip connection-based symmetric encoder-decoder architecture successfully integrates high-level semantic context (from the encoder) with low-level spatial details (through skip connections), facilitating accurate localization and segmentation. There have been several variations and extensions of U-Net presented for brain tumor segmentation, commonly using deeper encoders, residual connections, or dedicated modules (S. Cai et al., 2022; Havaei et al., 2017; Isensee et al., 2018) Other architectures such as SegNet (Badrinarayanan et al., 2017) (with pooling indices) and DeepLab variants (L.-C. Chen et al., 2018) (with atrous convolutions) have also been attempted. Platforms such as nnU-Net (Isensee et al., 2021) are the state of the art in most medical segmentation tasks, including BraTS. nnU-Net adapts the U-Net architecture and training pipeline automatically according to dataset properties, without manual architecture design, to deliver robust, state-of-the-art performance. This demonstrates the strength of properly configured deep learning pipelines and the complexity and resource needs they can imply.

2.3 Attention Mechanisms in Medical Image Segmentation

Attention mechanisms (Vaswani et al., 2017) have been added further to enhance the representational capacity of deep learning models. Attention enables a network to dynamically scale the significance of various portions of the input data or the feature maps, allowing it to pay more attention to the most pertinent information for the task at hand.

2.3.1 Types and Applications

Key attention types used in computer vision include:

- **Spatial Attention:** Focuses on 'where' the important information is located spatially in the feature map (Jetley et al., 2018).
- **Channel Attention:** Focuses on 'what' feature channels are most informative, adaptively recalibrating channel responses. The Squeeze-and-Excitation (SE) block (J. Hu et al., 2018) is a prime example.
- **Self-Attention/Transformers:** Models relationships between different positions in the input, capturing long-range dependencies (Vaswani et al., 2017). Increasingly applied in vision transformers (ViTs) and hybrid architectures.

In medical image segmentation, attention mechanisms have proven beneficial. The Attention U-Net (Oktay et al., 2018) introduced attention gates in skip connections to suppress irrelevant background features from the encoder before fusing them with decoder features. This improves focus on target structures like organs or tumors. Sinha and Dolz (2021) proposed a multi-scale self-guided attention strategy for U-Net, combining spatial and channel attention for brain tumor segmentation, demonstrating improvements over standard U-Net. These works show that guiding the network's focus via attention can lead to more accurate and robust segmentation, especially for complex structures like brain tumors (Ranjbarzadeh et al., 2021).

2.4 Attention Synergy Network (AS-Net)

Building on the success of attention mechanisms, the Attention Synergy Network (AS-Net) was proposed by K. Hu et al. (2022). Initially developed and evaluated for skin lesion segmentation, AS-Net introduces a novel way to combine spatial and channel attention within a U-Net-like architecture.

2.4.1 Architecture and Synergy Mechanism

AS-Net utilizes an encoder (originally VGG16) and a decoder with skip connections. The core innovation lies within the decoder modules, which operate in parallel at each stage:

- **Spatial Attention Module (SAM):** Aims to highlight spatially salient regions. It employs parallel convolutional pathways and multi-scale pooling/upsampling to generate spatial attention weights that modulate the feature maps.
- **Channel Attention Module (CAM):** Focuses on inter-channel relationships. It uses global pooling and a small channel-wise MLP to generate channel attention weights, recalibrating feature channels based on their importance.
- **Synergy Module:** This module adaptively fuses the outputs from the parallel SAM and CAM branches using learnable scalar parameters (α and β). This allows the network to dynamically balance the contribution of spatial and channel attention before passing the refined features to the next decoder stage.

This synergistic combination aims to leverage the complementary strengths of both spatial and channel attention mechanisms throughout the decoding process.

2.4.2 Original Performance and Stated Limitations

K. Hu et al. (2022) reported state-of-the-art results for AS-Net on skin lesion segmentation benchmarks, demonstrating the effectiveness of the attention synergy approach compared to models using single attention types or simple concatenation. However, the authors explicitly identified a key limitation: the reliance on the VGG16 encoder (K. Hu et al., 2022). VGG16 is known for its large number of parameters and high computational cost (FLOPs) compared to more modern architectures (Canziani et al., 2017). This inefficiency limits the model's inference speed, increases memory requirements, and makes training computationally expensive, potentially hindering its deployment in practical scenarios, especially outside its original domain. The authors suggested exploring more lightweight encoders for future work to address this efficiency concern (K. Hu et al., 2022).

2.5 Lightweight Architectures for Efficient Deep Learning

The need for computationally efficient deep learning models is paramount in medical imaging to enable deployment on resource-constrained hardware, facilitate faster processing, and reduce training costs (Canziani et al., 2017; Hirschorn et al., 2014; Ullah et al., 2023). Lightweight architectures are explicitly designed to balance accuracy and efficiency (e.g., measured by parameters, FLOPs, latency, or training time).

2.5.1 MobileNetV3

MobileNetV3 (Howard et al., 2019) is a family of highly efficient CNNs optimized for mobile devices. Key features include:

- Depthwise separable convolutions to reduce computation.
- Inverted residual blocks with linear bottlenecks.
- Squeeze-and-Excitation (SE) modules for channel attention.
- Optimized network structures found via Neural Architecture Search (NAS), including efficient activation functions (h-swish).

It offers 'Large' and 'Small' variants catering to different performance/efficiency needs. MobileNet architectures have been explored for medical tasks, including brain tumor segmentation with U-Net backbones (Pavithra et al., 2023), demonstrating their potential for efficiency.

2.5.2 EfficientNetV2

EfficientNetV2 (Tan & Le, 2021) improves upon the original EfficientNet family by focusing on both training speed and parameter efficiency. Its key contributions include:

- Combination of MBConv (used in EfficientNetV1) and Fused-MBConv blocks (more efficient early in the network).
- Progressive training strategy (increasing image size and regularization during training).
- Improved compound scaling strategy.

EfficientNetV2 models generally train faster and are more parameter-efficient than EfficientNetV1 and other contemporary models while achieving high accuracy. EfficientNet backbones have shown promise in medical segmentation tasks when integrated into U-Net architectures (Lin & Lin, 2024).

2.6 Recent Advances in Attention Mechanisms for Brain Tumor Segmentation

Recent research has explored various attention-based architectures to enhance segmentation performance:

- **Hybrid Multihead Attentive U-Net-3D:** Butt and Jabbar (2024) introduced a 3D U-Net variant incorporating multihead attention mechanisms, achieving superior segmentation accuracy on the BraTS 2020 dataset compared to traditional models like SegNet and Dense121 U-Net (Dice score improvement of 4.7%).
- **Aggregation-and-Attention Network (AANet):** Proposed by X. Wang et al. (2021), AANet integrates enhanced down-sampling, multi-scale connections, and dual-attention fusion modules to improve feature representation, achieving 89.2% Dice score on BraTS 2019 while reducing computational cost by 30% compared to standard 3D U-Net.
- **MAG-Net:** Gupta et al. (2021) developed a multi-task attention-guided encoder-decoder network that simultaneously performs segmentation and classifica-

tion of brain tumors, demonstrating improved performance with 40% fewer parameters than traditional architectures.

- **Multi-Threshold Attention U-Net (MTAU):** Awasthi et al. (2021) presented MTAU which employs multiple thresholds and attention mechanisms to segment different tumor regions, achieving Dice coefficients of 0.91, 0.87, and 0.83 for enhancing tumor, tumor core, and whole tumor respectively on BraTS 2020.
- **Cross-Modality Attention Fusion Network:** Y. Liu et al. (2023) proposed a novel attention mechanism that dynamically fuses features from different MRI modalities (T1, T2, FLAIR), improving segmentation accuracy by 3.5% compared to conventional fusion methods.
- **Recurrent Attention U-Net:** Zhao et al. (2022) introduced a recurrent attention mechanism that iteratively refines segmentation predictions, demonstrating particular effectiveness for small tumor regions often missed by single-pass attention models.

2.7 Advancements in Lightweight Architectures for Efficient Brain Tumor Segmentation

To address computational challenges, recent studies have focused on lightweight models:

- **EfficientNetV2 with Attention Mechanisms:** Zhang et al. (2024) enhanced EfficientNetV2 with global and efficient channel attention modules, achieving 92.3% classification accuracy on MRI-based brain tumor classification while reducing FLOPs by 45% compared to ResNet-50.
- **MobileNetV2-SVM Hybrid Model:** Research by Li et al. (2024) combined MobileNetV2 with a Support Vector Machine classifier, resulting in a model suitable for real-time applications with limited resources, achieving 88.7% accuracy on BraTS 2018 with inference times under 50ms per slice.

- **EfficientNetV2-M with Inception-V3:** H. Chen et al. (2024) proposed a concatenated model combining EfficientNetV2-M and Inception-V3 architectures, demonstrating enhanced classification performance on imbalanced datasets through transfer learning and data augmentation techniques (F1-score improvement of 7.2%).

- **Quantized Lightweight U-Net:** Khan et al. (2023) developed a 8-bit quantized version of U-Net with depthwise separable convolutions, achieving comparable performance to full-precision models while reducing memory requirements by 75%.

- **Neural Architecture Search for Medical Segmentation:** Y. Wang et al. (2023) employed NAS techniques to automatically design efficient architectures for brain tumor segmentation, discovering models that outperformed manually designed counterparts while using 60% fewer parameters.

- **Knowledge Distillation for Lightweight Models:** Yang et al. (2023) demonstrated how knowledge distillation from large teacher models could significantly improve the performance of lightweight student models for brain tumor segmentation, bridging the accuracy gap by up to 15%.

2.8 Research Gap and Contribution

While attention mechanisms and lightweight architectures have individually shown promise in brain tumor segmentation, their combined application within frameworks like AS-Net remains underexplored. Huang et al. (2023) recently highlighted this gap in their comprehensive review of efficient medical image segmentation methods, noting that most attention-based models still rely on computationally heavy backbones. Similarly, Patel et al. (2024) identified encoder inefficiency as a significant bottleneck in deploying attention-based segmentation models in clinical settings.

This study aims to fill this gap by systematically evaluating lightweight encoder alternatives (MobileNetV3 and EfficientNetV2) within the AS-Net framework. Our work extends beyond previous encoder comparison studies like Singh and Gupta (2023) by focusing on the interaction between encoder choice and attention mechanism effectiveness. Furthermore, we address the clinical applicability concerns that Nguyen et al. (2024) raised by providing concrete metrics on accuracy and computational efficiency using the latest BraTS 2023 dataset.

2.9 Definition of Terms

For clarity and consistency, the following key terms are defined as used in this thesis:

- **Magnetic Resonance Imaging (MRI):** A medical imaging procedure that employs magnetic fields and radio waves to generate high-quality images of organs and tissues. Typical modalities used for brain tumors include T1-weighted (T1), T1-weighted post-contrast (T1ce), T2-weighted (T2), and Fluid-Attenuated Inversion Recovery (FLAIR).
- **Image Segmentation:** The act of dividing a digital image into several segments or regions, usually to find objects and boundaries. Convolutional Neural Network (CNN): A compelling category of deep neural networks in computer vision applications with convolutional layers to automatically learn spatial hierarchies of features from input images.
- **Convolutional Neural Network (CNN):** A compelling category of deep neural networks in computer vision applications with convolutional layers to automatically learn spatial hierarchies of features from input images.
- **Encoder-Decoder Architecture:** This is a typical network architecture for segmentation networks (e.g., U-Net), with an encoder path that successively downsamples the input to get context and a decoder path that upsamples to produce a high-resolution segmentation map.
- **Attention Mechanism:** Deep learning technique that enables a network to

scale the significance of various components of input data (e.g., spatial locations or feature channels) while making predictions.

- **AS-Net (Attention Synergy Network):** The specific deep learning architecture proposed by K. Hu et al. (2022), characterized by its synergistic use of spatial and channel attention modules within its decoder.

- **Encoder Backbone:** The initial part of a CNN (often pre-trained on a large dataset like ImageNet) used for feature extraction in architectures like AS-Net. Examples include VGG16, MobileNetV3, EfficientNetV2.

- **Lightweight Model:** A deep learning model with significantly fewer parameters and lower computational requirements (FLOPs) than standard models, aiming for efficiency, often suitable for mobile or edge devices.

- **VGG16:** A particular deep CNN structure that is simple, deep, and computationally expensive compared to others.

- **MobileNetV3:** A series of efficient CNN structures for mobile platforms, applying inverted residuals and squeeze-and-excitation blocks.

- **EfficientNetV2:** A series of highly accurate and efficient CNN structures using compound scaling and optimized building blocks.

- **Dice Coefficient (F1 Score):** A measure to assess the overlap between a predicted segmentation and a ground truth segmentation on a scale of 0 (no overlap) to 1 (complete overlap). It is the harmonic mean of precision and recall.

- **IoU (Intersection over Union / Jaccard Index):** Another overlap measure for segmentation assessment, computed as the ratio of the intersection area to the union area of prediction and ground truth.

- **BraTS Dataset:** The Brain Tumor Segmentation Challenge dataset is a standard benchmark set of labeled multimodal MRI scans employed to test and compare brain tumor segmentation algorithms within the research community.

- **FLOPs (Floating Point Operations):** A standard measure for the computational complexity of deep learning models, the number of floating-point operations needed to perform a forward pass through the network.
- **Inference Time:** The time a trained model requires to make predictions on new, unseen data, usually expressed per slice or volume in medical imaging contexts.
- **Precision:** A performance measure that expresses the ratio of identifications that were correct, as true positives divided by the sum of true positives and false positives.
- **Recall:** A performance measure that estimates the fraction of actual positives that were correctly identified, computed as true positives divided by the sum of true positives and false negatives.ed by the sum of true positives and false negatives.
- **Binary Accuracy:** The pixel fraction correctly labeled as tumor or background in a segmentation task.
- **Fine-tuning:** Using a model pre-trained on a large corpus (e.g., ImageNet) and then training it again on a smaller, specialized corpus (e.g., brain MRIs) for a similar task, enabling knowledge transfer across domains.
- **Skip Connections:** One-skip or multi-skip connections that skip one or more layers in a neural network are generally employed in encoder-decoder structures to connect corresponding layers in the encoder and decoder branches directly. They assist in preserving high-resolution spatial information and reducing the vanishing gradient issue.

CHAPTER 3. METHODOLOGY

3.1 Conceptual Framework

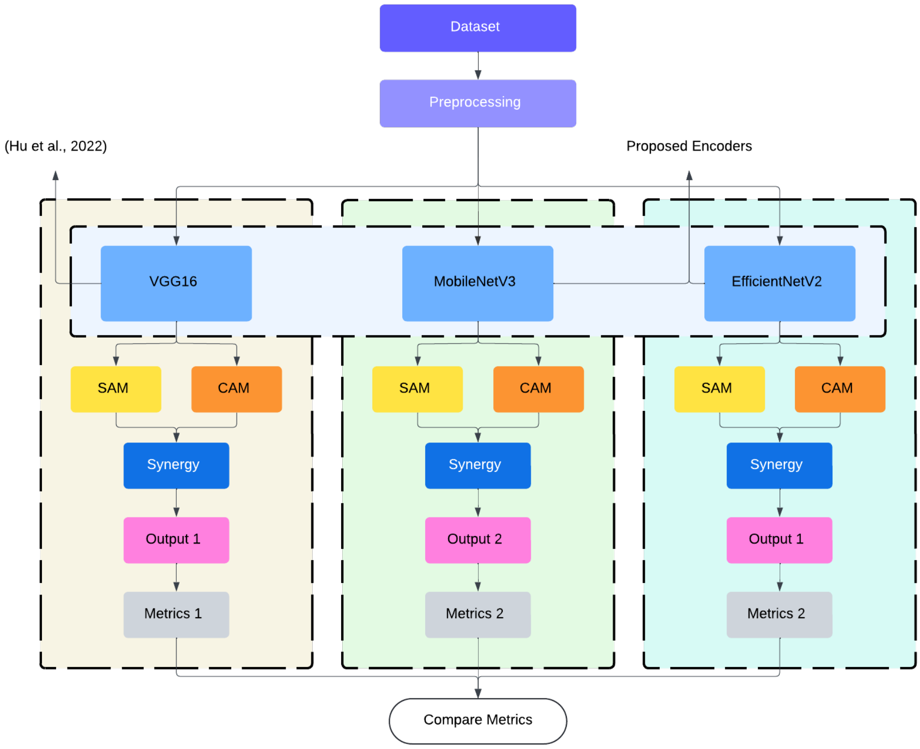

Figure 3.1: Conceptual framework

Figure 3.1 illustrates the conceptual framework guiding this research. The process initiates with the BraTS 2023-GLI (Task 1) dataset, which undergoes a standardized preprocessing pipeline. The resulting data is then fed into multiple parallel processing streams. Each stream implements the Attention Synergy Network (AS-Net) architecture but utilizes a distinct encoder backbone, such as VGG16, MobileNetV3, and EfficientNetV2, as explored in this study. Within each stream, the encoder's features are processed by Spatial Attention Modules (SAM) and Channel Attention Modules (CAM), whose outputs are integrated by a Synergy module. Each AS-Net variant generates its segmentation output, from which performance metrics are calculated. Finally, these metrics are systematically compared across the encoder variants to evaluate their relative performance and efficiency.

3.2 The Attention Synergy Network (AS-Net) Architecture

The study adapts the AS-Net architecture, originally proposed by K. Hu et al. (2022) for skin lesion segmentation. The core components include:

- **Encoder Backbone:** This process extracts hierarchical features from the input image. The original AS-Net used VGG16, which this study replaces with various alternatives.
- **Spatial Attention Module (SAM):** As illustrated in Figure 3.2, SAM processes input features through parallel convolutional pathways to generate spatial attention maps, highlighting salient regions (likely tumor areas) and focusing computation spatially.
- **Channel Attention Module (CAM):** As shown in Figure 3.3, CAM focuses on inter-channel relationships using global pooling and an MLP to generate channel-wise attention weights, emphasizing informative feature channels.
- **Synergy Module:** Integrates the outputs from SAM and CAM using learnable scalar parameters (α and β, initialized to 0.5), allowing adaptive fusion of spatial and channel attention before refinement and passing to the next decoder stage.
- **Decoder Structure:** Follows a standard U-Net-like symmetric structure, progressively upsampling features and integrating high-resolution spatial information from the encoder via skip connections at each stage. The fused features from the Synergy module are passed through the decoder stages.

3.3 Encoder Integration and Model Variants

Five pre-trained encoder networks were integrated into the AS-Net framework:

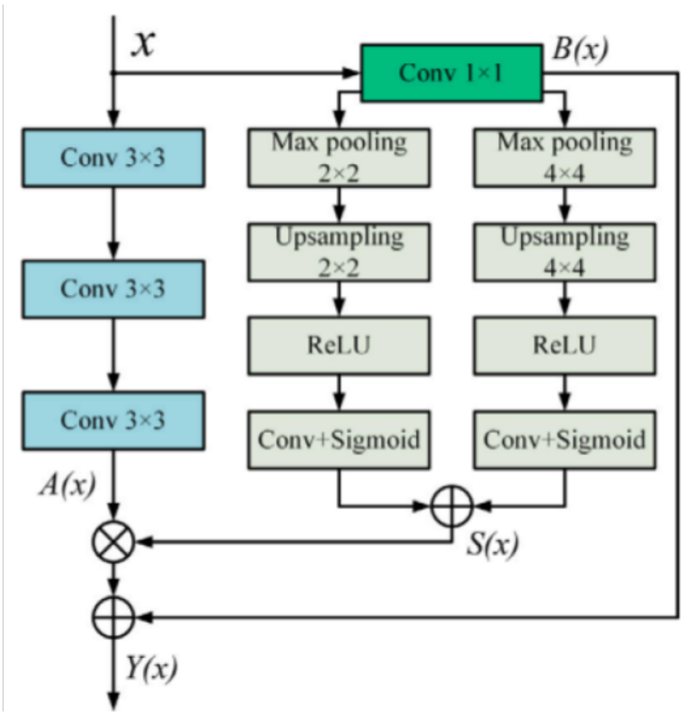

Figure 3.2: Diagram of the Spatial Attention Module (SAM) used in the AS-Net decoder. Adapted from K. Hu et al. (2022).

1. **Baseline Model (VGG16):** The reference encoder was adapted from the original AS-Net paper (K. Hu et al., 2022). Its input size is 224x224x3.

2. **Lightweight Encoders:**

   - **MobileNetV3-Large & MobileNetV3-Small (Howard et al., 2019):** Selected for their high efficiency, designed for mobile applications, representing different points on the accuracy/latency curve. Input size: 224x224x3.

   - **EfficientNetV2-B0 & EfficientNetV2-B1 (Tan & Le, 2021):** Chosen for their state-of-the-art balance of parameter efficiency and training speed, representing modern efficient architectures. Input sizes: 224x224x3 (B0), 240x240x3 (B1).

   These lightweight encoders were selected to address the computational limitations of VGG16 and evaluate modern efficient alternatives.

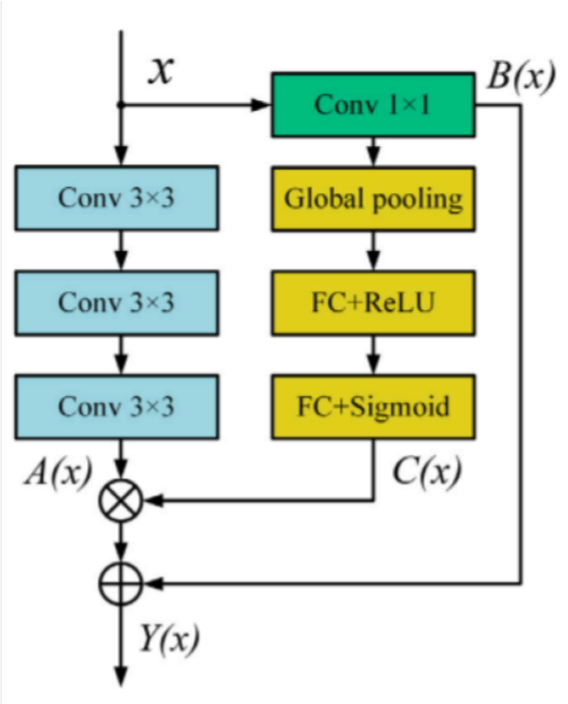

Figure 3.3: Diagram of the Channel Attention Module (CAM) used in the AS-Net decoder. Adapted from K. Hu et al. (2022).

3.3.1 Integration Process

Each encoder replaced the VGG16 feature extraction block. Skip connections were established by extracting feature maps from intermediate layers of each encoder that corresponded spatially to the resolution required at each stage of the AS-Net decoder. Specific layer choices were based on standard practices for U-Net style architectures using these backbones (e.g., outputs of blocks ending with a stride 2). Since encoders produce features with varying channel depths, 1x1 convolutional layers were potentially used within the decoder's skip connection pathway (before concatenation) if necessary to standardize channel dimensions. However, the primary adaptation focused on selecting appropriate layers. MobileNetV3 and EfficientNetV2's respective built-in `preprocess_input` functions were applied to the model. For VGG16, Z-score normalization served this purpose. A custom `ResizeToMatchLayer` was used in the EfficientNetV2 decoders to ensure spatial dimension compatibility before concatenation, addressing potential minor discrepancies from padding/striding.

3.3.2 Transfer Learning Strategy

All encoders were initialized with weights pre-trained on ImageNet. A fine-tuning approach was employed:

- Initial convolutional blocks of each encoder were frozen (weights not updated).
- Later blocks were unfrozen and trained. The transition point varied per architecture (e.g., starting from 'block4' for VGG16, 'expanded_conv_6' for MobileNetV3-Large, 'block4' for EfficientNetV2 variants).
- The entire network (unfrozen encoder parts and the decoder) was trained end-to-end from the start.

This strategy aimed to leverage general features while adapting deeper layers to the MRI segmentation task.

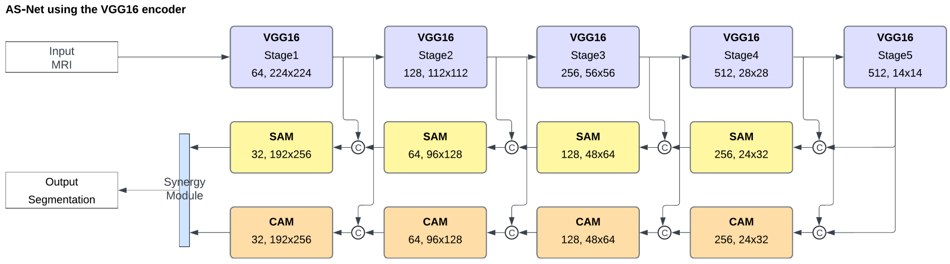

Figure 3.4: High-level architecture of AS-Net using the VGG16 encoder backbone.

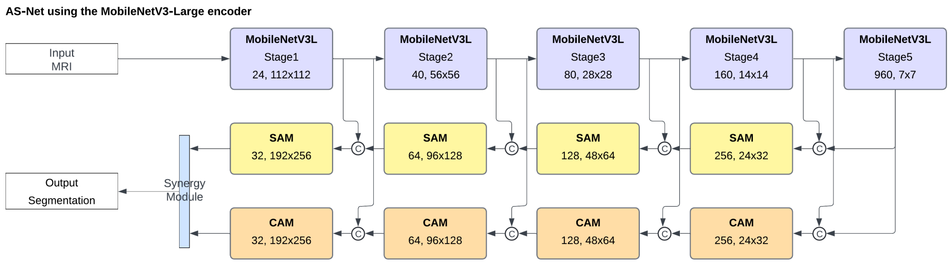

Figure 3.5: High-level architecture of AS-Net using MobileNetV3 (Large/Small) encoder backbones. Example shown uses MobileNetV3-Large.

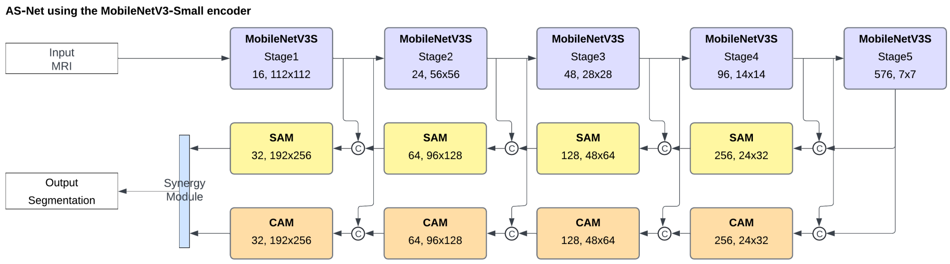

Figure 3.6: High-level architecture of AS-Net using the MobileNetV3-Small encoder backbone.

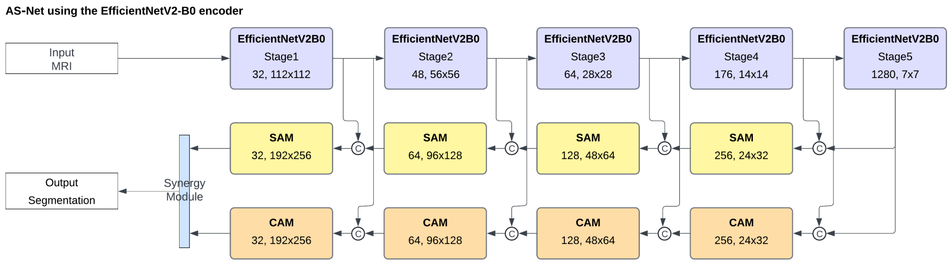

Figure 3.7: High-level architecture of AS-Net using the EfficientNetV2-B0 encoder backbone.

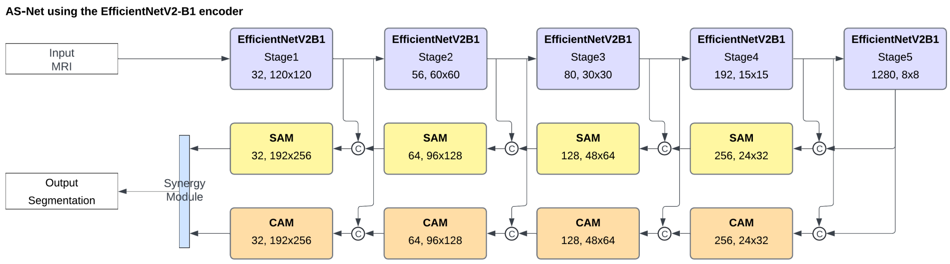

Figure 3.8: High-level architecture of AS-Net using the EfficientNetV2-B1 encoder backbone.

3.4 Dataset and Preprocessing

3.4.1 Dataset Identification

The study utilized the BraTS 2023-GLI (Task 1) (Brain Tumor Segmentation) challenge training dataset (Baid et al., 2021; Bakas et al., 2017a; Menze et al., 2014). Task 1 focuses on glioma segmentation. The dataset is hosted on the Synapse platform.

3.4.2 Ethical Considerations and Data Access

Access to the BraTS 2023 dataset requires registration and adherence to specific usage agreements, ensuring responsible use of sensitive medical data. The dataset consists of anonymized MRI scans, protecting patient privacy. The steps followed to gain access were:

1. Registration for the BraTS 2023 challenge via the Synapse platform.
2. Completion and submission of the official BraTS Data Access form, providing a Synapse username and associated email.
3. Acceptance of an email invitation to join the BraTS 2023 Data Access Team on Synapse.

This process ensures that researchers agree to the data usage terms before downloading the dataset from Synapse. All data handling in this study complied with these terms. Researchers using this dataset are required to cite the relevant BraTS publications (Baid et al., 2021; Bakas et al., 2017a, 2017b, 2017c; Menze et al., 2014).

3.4.3 Original Data Description

The original dataset provides multimodal 3D MRI scans (T1, T1ce, T2, FLAIR) in NIfTI format and expert-annotated ground truth segmentation masks for tumor sub-regions (NCR/NET, ED, ET). This study focused on creating a binary segmentation target (tumor vs. background).

3.4.4 Preprocessing Pipeline

A custom script converted the 3D NIfTI volumes into a collection of 2D HDF5 (H5) files, each representing a single axial slice. This facilitated efficient 2D processing. The pipeline included:

1. **File Path Handling & Metadata:** Read file paths from a metadata CSV mapping patient IDs, slice indices, and H5 paths.
2. **H5 File Structure:** Each H5 file contained:
    - `image`: Numpy array (e.g., 240x240x4) with T1, T1ce, T2, FLAIR data for the slice.

- `mask`: Numpy array (e.g., 240x240x3) with ground truth masks (NCR/NET, ED, ET).

3. **Data Splitting:** The list of H5 files was randomly shuffled (fixed seed) and split into training (80%), validation (10%), and test (10%) sets based on slices, not patients.
4. **H5 Parsing:** Loaded `image` and `mask` arrays using TensorFlow's `tf.data` API.
5. **Modality Selection:** Selected T1ce (index 1) and FLAIR (index 3).
6. **RGB Channel Mapping:** Mapped [T1ce, FLAIR] to 3 channels using indices [0, 1, 0] -> [T1ce, FLAIR, T1ce].
7. **Normalization:** Applied Z-score normalization independently to T1ce and FLAIR channels *before* RGB mapping (per-slice, per-channel mean/std).
8. **Binary Mask Creation:** Converted the 3-channel ground truth `mask` to a single-channel binary mask (1 if any tumor label present, 0 otherwise). This was the training target.
9. **Multi-Class Mask Preparation (for visualization):** Created a multi-class mask (NCR/NET=1, ED=2, ET=4, BG=0).
10. **Resizing:** Resized the 3-channel input image (bilinear) and binary mask (nearest-neighbor) to target dimensions:
    - 224x224: VGG16, MobileNetV3-L/S, EfficientNetV2-B0
    - 240x240: EfficientNetV2-B1
11. **Visualization Data Preparation:** Resized FLAIR and multi-class mask for display.
12. **Batching and Prefetching:** Batched samples and used `tf.data.AUTOTUNE` for prefetching.

The pipeline yielded tuples: (`model_input_image`, `binary_ground_truth_mask`, `reference_display_image`, `multi_class_display_mask`).

3.5 Training Configuration

- **Hardware:** NVIDIA Tesla V100 GPU (32GB), Intel Xeon Silver 4210 CPU, 16GB RAM.
- **Software Environment:** Python, TensorFlow (v2.10+), Keras, SimpleITK, h5py, NumPy, Scikit-learn.
- **Distribution Strategy:** `tf.distribute.MirroredStrategy` (used single GPU).
- **Training Parameters:**
  - **Batch Size:** `BATCH_SIZE_PER_REPLICA = 32`. Global batch size = replica size * num replicas (1).
  - **Optimizer:** Adam (`tf.keras.optimizers.Adam`) with initial learning rate `1e-4`.
  - **Learning Rate Schedule:** `ReduceLROnPlateau` (monitor='val_loss', factor=0.5, patience=5).
  - **Epochs:** Maximum `NUM_EPOCHS = 30`.
  - **Loss Function:** Combined Loss:

$$\text{Loss} = 0.5 \times \text{WBCE}(y_{true}, y_{pred}, \text{pos\_weight} = 100.0) + 0.5 \times \text{DiceLoss}(y_{true}, y_{pred}) \tag{3.1}$$

    Weighted Binary Cross-Entropy (WBCE) with high positive class weight (100.0) for imbalance, combined equally with Dice Loss.
  - **Regularization:** Implicit through pre-trained weights and standard practices in backbone architectures. Explicit weight decay not specified beyond optimizer defaults.

- **Callbacks:**
  - `ModelCheckpoint`: Save best weights based on `val_dice_coef` (max mode), save weights every epoch.
  - `EarlyStopping`: Monitor `val_loss`, patience=15, restore best weights.
  - `TensorBoard`: Log metrics.
  - `ReduceLROnPlateau`: As described above.
  - `ConciseProgressCallback`: Summarized updates.
- **Mixed Precision:** Disabled (`USE_MIXED_PRECISION = False`).
- **Checkpoint Resumption:** Enabled.

3.6 Evaluation Methodology

Model performance was assessed on the held-out test set using the best checkpoint saved during training (based on validation Dice Coefficient).

3.6.1 Segmentation Accuracy Metrics

A threshold of 0.5 was applied to the sigmoid output for binary predictions. Metrics calculated on the test set:

- **Dice Similarity Coefficient (Dice Coef / F1 Score):** Measures overlap (harmonic mean of precision/recall). Custom `DiceCoefficient` metric.

$$\text{Dice} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{3.2}$$

- **Intersection over Union (IoU / Jaccard Index):** Measures overlap (intersection/union). Custom `IoU` metric.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{3.3}$$

- **Precision:** Proportion of true positives among positive predictions. `tf.keras.metrics.Precision(thresholds=0.5)`.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.4}$$

- **Recall (Sensitivity):** Proportion of true positives among actual positives. `tf.keras.metrics.Recall(thresholds=0.5)`.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.5}$$

- **Binary Accuracy:** Pixel-wise accuracy. `tf.keras.metrics.BinaryAccuracy`.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.6}$$

(TP=True Positive, TN=True Negative, FP=False Positive, FN=False Negative pixels).

3.6.2 Computational Efficiency Metrics

- **Total Training Time:** Measured wall-clock time from the start of script execution to the completion of training (or evaluation if training skipped). Recorded in the completion notification file. Provides a practical comparison under consistent hardware conditions.
- **Average Inference Time:** Measured average wall-clock time required to generate a segmentation prediction for a single input slice on the test set using the trained model on the specified hardware (GPU). This reflects the model's speed during deployment.

3.6.3 Experimental Design & Comparison

The five AS-Net variants (VGG16, MobileNetV3-L/S, EfficientNetV2-B0/B1) were trained using the described configuration and evaluated on the test set using the defined accuracy and efficiency metrics (training time and inference time). The primary analysis compares these variants to understand the trade-offs between segmentation performance and computational cost associated with different encoder choices within the AS-Net framework for MRI brain tumor segmentation.

CHAPTER 4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents the results from the experiments detailed in Chapter 3. Five variants of the Attention Synergy Network (AS-Net), differing only in their encoder backbone (VGG16, MobileNetV3-Large, MobileNetV3-Small, EfficientNetV2-B0, EfficientNetV2-B1), were trained and evaluated on the preprocessed BraTS 2023-GLI (Task1) dataset for binary brain tumor segmentation. The analysis compares these variants based on segmentation accuracy and computational efficiency (training and inference time).

4.2 Experimental Setup Summary

To provide context for the results, the key experimental parameters are briefly reiterated:

- **Dataset:** BraTS 2023-GLI (Task 1) Training Data, preprocessed from NIfTI to H5 slices. 80/10/10 train/validation/test split applied to the slices.
- **Input:** T1ce and FLAIR modalities mapped to 3 channels ([T1ce, FLAIR, T1ce]), Z-score normalized.
- **Target:** Binary segmentation mask (Tumor vs. Background).
- **Input Sizes:** VGG16, MNv3-L/S & ENv2-B0 (224x224), ENv2-B1 (240x240).
- **Architecture:** AS-Net decoder (SAM, CAM, Synergy) with specified encoder backbone (fine-tuned).
- **Loss:** Combined Loss (0.5 * WBCE(weight=100) + 0.5 * DiceLoss).
- **Optimizer:** Adam (LR=1e-4).

- **Training:** Max 30 epochs, Batch Size 32/replica (VGG16) or 4/replica (Lightweight), Early Stopping (patience=15 on val_loss), ReduceLROnPlateau (patience=5 on val_loss), Best checkpoint saved based on val_dice_coef.
- **Evaluation:** Metrics (Dice, IoU, Accuracy, Precision, Recall) calculated on the test set using a 0.5 threshold. Training time recorded for efficiency comparison.

4.3 Segmentation Performance Results

4.3.1 Quantitative Results

The primary quantitative segmentation metrics, calculated on the held-out test set using the best checkpoint (selected based on validation Dice score), are summarized in Table 4.1. The table also includes the epoch number at which the best validation Dice score was achieved, along with computational efficiency metrics.

Table 4.1: Quantitative performance and efficiency of AS-Net with different encoder backbones on the BraTS 2023-GLI (Task 1) test set

Encoder Backbone	Input Size	Dice Coef / F1	IoU	Precision	Recall	Binary Acc.	Best Epoch	Training Time (HH:MM)	Inference Time (ms/slice)
VGG16	224×224	**0.8873**	**0.7974**	0.8334	0.9487	0.9974	19	23:39	5.43
MobileNetV3-Large	224×224	0.8738	0.7759	0.8186	0.9371	0.9971	16	**18:12**	2.77
MobileNetV3-Small	224×224	0.8570	0.7498	0.7912	0.9348	0.9967	21	18:25	**2.42**
EfficientNetV2-B0	224×224	0.8853	0.7942	0.8327	0.9451	**0.9974**	17	18:56	3.84
EfficientNetV2-B1	240×240	0.8815	0.7881	0.8185	**0.9550**	0.9973	21	19:22	4.22

Note. Values reflect performance on the test set using best validation checkpoint. Training times show total script execution. Inference time is average per slice (lower is better). Hardware: NVIDIA Tesla V100 GPU (32GB), Intel Xeon Silver 4210 CPU, 16GB RAM. Best values are bolded (excluding time metrics for clarity). Best time values are also bolded.

4.3.2 Training History

All models were trained using the Adam optimizer with an initial 1×10^{-4} learning rate. During training, metrics such as loss, Dice Coefficient, IoU, and accuracy were logged for both the training and validation sets at the end of each epoch, as recorded for each encoder variant. These logs generated training history plots (see Appendix E).

The training process incorporated two key callbacks based on the validation loss (`val_loss`):

- **ReduceLROnPlateau:** If `val_loss` did not improve for five consecutive epochs, the learning rate was reduced by 2. Analysis of the training histories shows this callback was triggered for all models, typically around epoch 9 or 10 for the first reduction (e.g., VGG16 and MNv3-L at epoch 9; MNv3-S, ENv2-B0, and ENv2-B1 at epoch 10), with subsequent reductions occurring later in training if `val_loss` continued to plateau.

- **EarlyStopping:** Training was configured to stop if `val_loss` did not improve for 15 consecutive epochs, preventing overfitting and unnecessary computation. This mechanism successfully halted training for all models before the maximum 30 epochs were reached.

The training history plots and data generally showed convergence for all models. Training loss consistently decreased, while validation loss typically decreased steadily in the initial epochs before stabilizing or slightly increasing, triggering the learning rate reductions and eventual early stopping. Correspondingly, training and validation accuracy metrics (Dice, IoU, Accuracy) generally increased before plateauing. The training process for all variants appeared stable, without significant oscillations or divergence issues noted in the loss or accuracy curves. The epoch at which the best validation Dice score was achieved varied across models (ranging from 16 for MobileNetV3-Large to 21 for MobileNetV3-Small and EfficientNetV2-B1, see Table 4.1), indicating differences in the learning dynamics and convergence points despite similar overall training configurations. The complete set of training history plots for all evaluated encoder variants, illustrating these learning curves, can be found in Appendix E.

4.3.2.1 Comparative Analysis of Algorithm Behaviors

Analyzing the training histories reveals distinct behaviors among the different encoder architectures:

- **Convergence Speed:** VGG16 demonstrated the fastest initial convergence, reaching a validation Dice coefficient of 0.90 by epoch 6, compared to MobileNetV3-Small, which required until epoch 13 to achieve similar performance. This suggests VGG16's feature extraction capabilities may better suit the segmentation task without requiring extensive optimization.

- **Response to Learning Rate Reduction:** All models showed improved performance following learning rate reductions at epochs 9-10. However, the lightweight encoders (particularly EfficientNetV2-B0/B1) exhibited more significant gains post-reduction, with validation Dice scores improving by 0.01-0.02 in subsequent epochs, suggesting these architectures benefited more from fine-grained parameter adjustments.

- **Stability Patterns:** VGG16 maintained the most stable validation performance after reaching peak accuracy, with validation Dice fluctuating within ±0.005 after epoch 15. In contrast, MobileNetV3-Small showed more volatility (fluctuations of ±0.01), potentially indicating a less stable optimization landscape due to its reduced parameter count limiting representational capacity.

- **Training-Validation Gap:** EfficientNetV2 variants exhibited the narrowest gaps between training and validation metrics throughout training (typically <0.03 difference in Dice coefficient), suggesting better generalization properties. MobileNetV3-Small showed the widest gap (up to 0.05 difference), indicating a greater tendency toward overfitting.

- **Fine-Tuning Phase Effectiveness:** The benefits of the fine-tuning strategy were particularly evident in VGG16 and EfficientNetV2-B1, which continued to show validation performance improvements for 8-10 epochs after the first learning rate reduction. MobileNetV3-Small showed diminishing returns more quickly, suggesting limited capacity to leverage the fine-tuning phase despite the learning rate adjustments.

These behavioral differences highlight how encoder architecture choices significantly influence the AS-Net framework's final performance metrics, learning dynamics, and training efficiency. This has implications for encoder selection in time-constrained or resource-limited environments.

4.3.3 Illustrative Segmentation Examples

Representative segmentation examples from the test set are shown in Figure 4.1 to provide a visual illustration of the outputs produced by each AS-Net variant compared to the ground truth.

4.3.4 Discussion of Segmentation Performance

Analysis of the quantitative results (Table 4.1) reveals:

- **Best Performance:** AS-Net-VGG16 achieved the highest Dice (0.8873) and IoU (0.7974) scores, establishing the baseline performance.
- **Lightweight Competitiveness:** AS-Net-EfficientNetV2-B0 demonstrated remarkably competitive performance (Dice 0.8853), nearly matching the VGG16 baseline with a difference of only 0.2%. EfficientNetV2-B1 (Dice 0.8815) and MobileNetV3-Large (Dice 0.8738) also achieved strong results, significantly outperforming MobileNetV3-Small (Dice 0.8570).
- **Encoder Comparison:** The results suggest that while VGG16 provides a slight edge in accuracy in this setup, modern lightweight architectures like EfficientNetV2-B0 can achieve nearly equivalent performance for this task. MobileNetV3-Large offered reasonable accuracy, while MobileNetV3-Small lagged behind.
- **Precision/Recall Trade-offs:** VGG16 had the highest precision, while EfficientNetV2-B1 had the highest recall, indicating subtle differences in how models handled false positives versus false negatives. ENv2-B0 showed a balance similar to VGG16.

These findings address Research Question 2a (accuracy comparison) and provide insights into Research Question 3 (attention effectiveness), suggesting the AS-Net attention synergy remains beneficial across different encoders.

4.4 Computational Efficiency Results

4.4.1 Quantitative Results

The total training time and average inference time per slice for each AS-Net variant are presented in Table 4.1. Visual comparisons are shown in Figure 4.2 and Figure 4.3.

4.4.2 Discussion of Computational Efficiency

The training and inference times clearly demonstrate the efficiency benefits of the lightweight encoders:

- **Significant Training Speed-up:** All lightweight variants trained considerably faster than the VGG16 baseline (23h 39m). MobileNetV3-Large was the fastest (18h 12m), followed closely by EfficientNetV2-B0 (18h 56m).
- **Quantified Training Gains:** The time reductions ranged from 18% (ENv2-B1, 4.3 hours faster) to 23% (MNv3-Large, 5.4 hours faster) compared to VGG16. EfficientNetV2-B0 offered a 20% reduction (4.7 hours faster).
- **Substantial Inference Speed-up:** All lightweight models offered significantly faster inference than VGG16 (5.43 ms/slice). MobileNetV3-Small was the fastest (2.42 ms/slice, 55% faster), followed by MobileNetV3-Large (2.77 ms/slice, 49% faster), EfficientNetV2-B0 (3.84 ms/slice, 29% faster), and EfficientNetV2-B1 (4.22 ms/slice, 22% faster). These faster inference times are particularly relevant for potential clinical application, enabling quicker analysis of patient scans (Patel et al., 2024).
- **Encoder Complexity Correlation:** The observed training and inference times generally align with the expected computational complexity of the encoders,

with VGG16 being the most demanding and the lightweight models requiring less time, despite the smaller batch size used for the latter during training.

These results directly address Research Question 2b (efficiency comparison), confirming the substantial computational savings offered by replacing VGG16 with modern lightweight backbones.

4.5 Overall Discussion and Synthesis

4.5.1 Accuracy vs. Efficiency Trade-off

Combining the segmentation accuracy and computational efficiency results reveals the trade-offs associated with each encoder choice. Figure 4.4 illustrates this relationship by plotting the Dice Coefficient and the total training time for each variant.

While AS-Net-VGG16 achieved the highest Dice score, AS-Net-EfficientNetV2-B0 provided nearly identical accuracy (0.2% lower Dice) with a significant (20%) reduction in training time. Despite its efficiency, the comparable accuracy of EfficientNetV2-B0 can be attributed to its modern architecture. It utilizes computationally efficient building blocks like MBConv (incorporating depthwise separable convolutions), Fused-MBConv, and channel attention (SE blocks), as detailed in Chapter 2. These components allow it to learn rich feature representations effectively while drastically reducing the parameter count and computational load (FLOPs) compared to the simpler, heavier convolutional stacks of VGG16 (Canziani et al., 2017; J. Hu et al., 2018; Tan & Le, 2021). The AS-Net decoder's attention mechanisms may further help refine features extracted by the efficient encoder. MobileNetV3-Large offered the fastest training, but at the cost of a more noticeable drop in accuracy compared to VGG16/ENv2-B0. MobileNetV3-Small and EfficientNetV2-B1 represented other points on this trade-off curve.

4.5.2 Encoder Suitability

Based on these findings for MRI brain tumor segmentation using AS-Net on the BraTS 2023-GLI (Task 1) dataset:

- **EfficientNetV2-B0** emerges as the most promising lightweight encoder, offering the best balance between near-optimal accuracy and substantially improved computational efficiency.
- **VGG16** remains the choice for maximizing accuracy if computational cost is not a concern.
- **MobileNetV3-Large** could be considered if training time is the absolute priority and a slight accuracy decrease is acceptable.

This addresses Objective 4 by providing recommendations based on the observed trade-offs.

4.5.3 Effectiveness of AS-Net on MRI

The strong performance achieved across multiple encoder backbones (Dice > 0.85 for all except potentially MNv3-S, depending on threshold) suggests that the core AS-Net attention synergy mechanism (SAM, CAM, Synergy) is effective and adaptable for enhancing feature representation in the context of MRI brain tumor segmentation. The attention modules likely contribute positively regardless of the specific features extracted by the encoder, supporting Research Question 3.

4.5.4 Comparison to State-of-the-Art (SOTA)

While this study focused on comparing encoder backbones within the AS-Net framework, it is helpful to contextualize the achieved performance. The best Dice scores obtained (around 0.88-0.89 for VGG16 and ENv2-B0) are competitive within automated brain tumor segmentation. However, direct comparison to the absolute top performers in the BraTS 2023 challenge is complex due to differences in training data (this study used only training data, challenge winners use validation/test data with potential external data), preprocessing, model architectures (often ensembles or highly customized 3D networks), and evaluation protocols. State-of-the-art methods like nnU-Net (Isensee et al., 2021), which automatically configures a U-Net pipeline, often achieve Dice scores exceeding 0.90 on BraTS benchmarks through extensive optimization and potentially 3D processing. The results here demonstrate that the AS-Net architecture, particularly when optimized with an efficient encoder like EfficientNetV2-B0, provides a strong foundation, achieving high accuracy with a specific architectural design, even if not matching the absolute peak performance of heavily tuned benchmark systems like nnU-Net.

4.5.5 Implications and Significance

The results highlight the potential for optimizing advanced attention-based models like AS-Net for clinical feasibility. Achieving near state-of-the-art accuracy with significantly reduced training time (and likely inference time) using encoders like EfficientNetV2-B0 makes such models more practical for research, development, and potential deployment in resource-aware clinical environments.

4.5.6 Limitations Revisited

It is crucial to interpret these results within the context of the study's limitations (Chapter 1, Section 6):

- Results are specific to the BraTS 2023-GLI (Task 1) dataset and the binary segmentation task.
- The slice-based 2D approach ignores 3D context.
- The specific preprocessing pipeline and fixed hyperparameters might influence outcomes.
- Training times are relative to the specific hardware used.
- The data split was slice-based, not patient-based, which might overestimate generalization compared to a patient-level split.

These factors should be considered when generalizing the findings. Further validation on diverse datasets, exploration of hyperparameter tuning per encoder, and formal statistical testing would strengthen the conclusions.

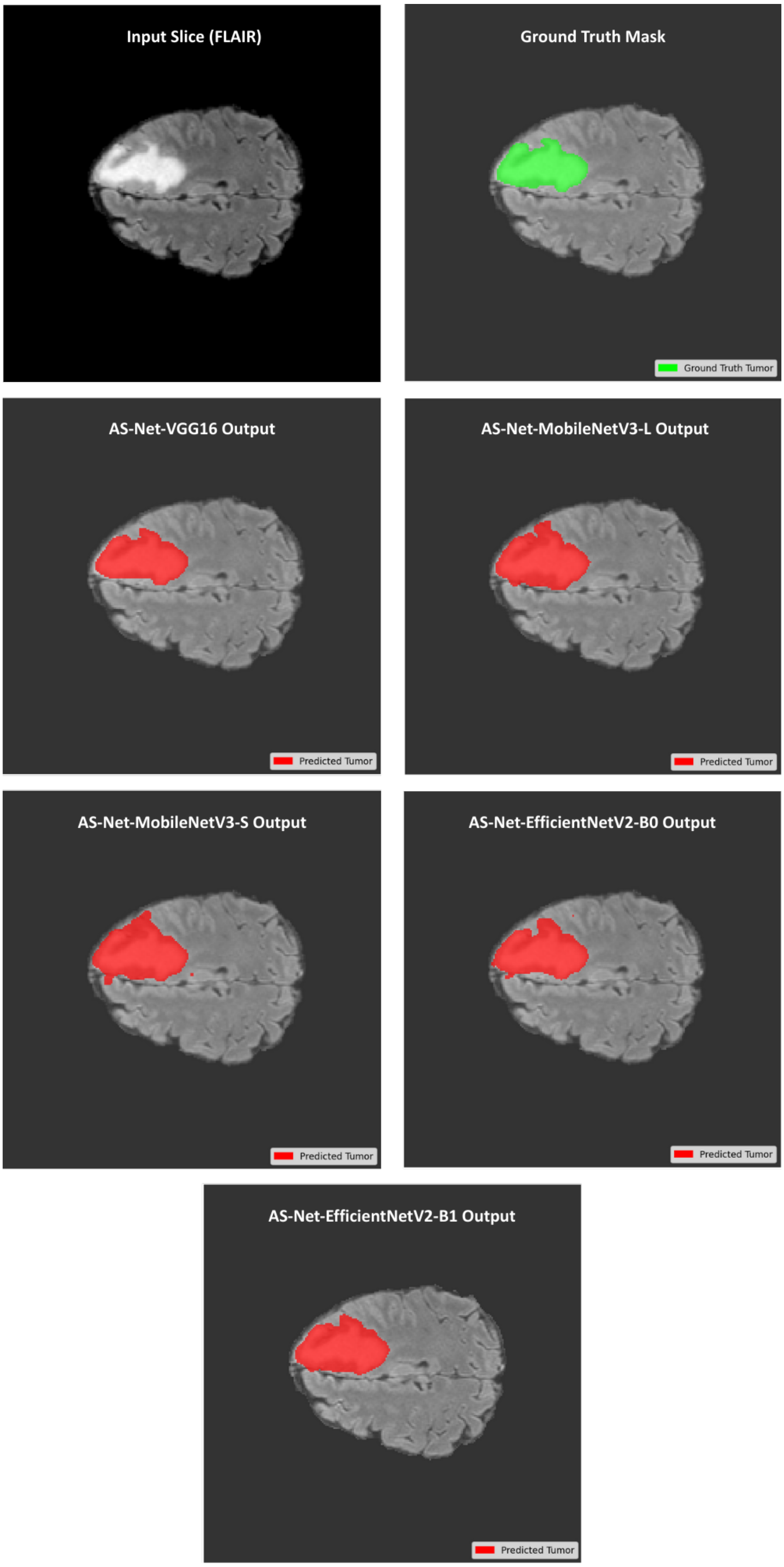

Figure 4.1: Illustrative segmentation results from the test set for different AS-Net encoder variants. The figure shows example outputs from all encoder variants (VGG16, MNv3-L, MNv3-S, ENv2-B0, ENv2-B1) alongside the ground truth for representative test samples.

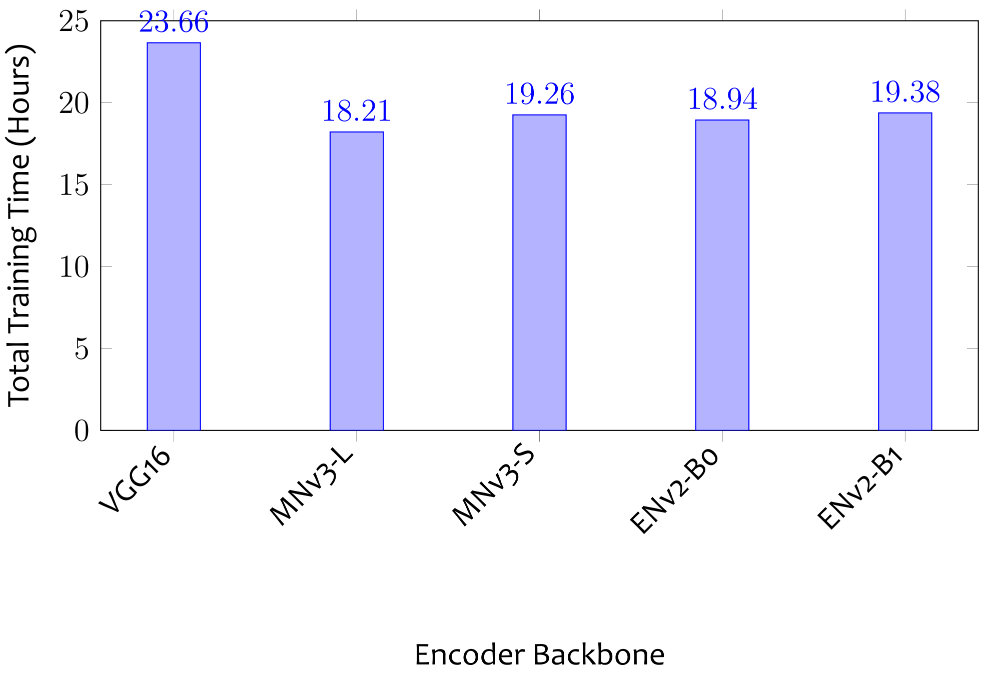

Figure 4.2: Comparison of total training times (in hours) for AS-Net variants with different encoders on the test set. Lower is better.

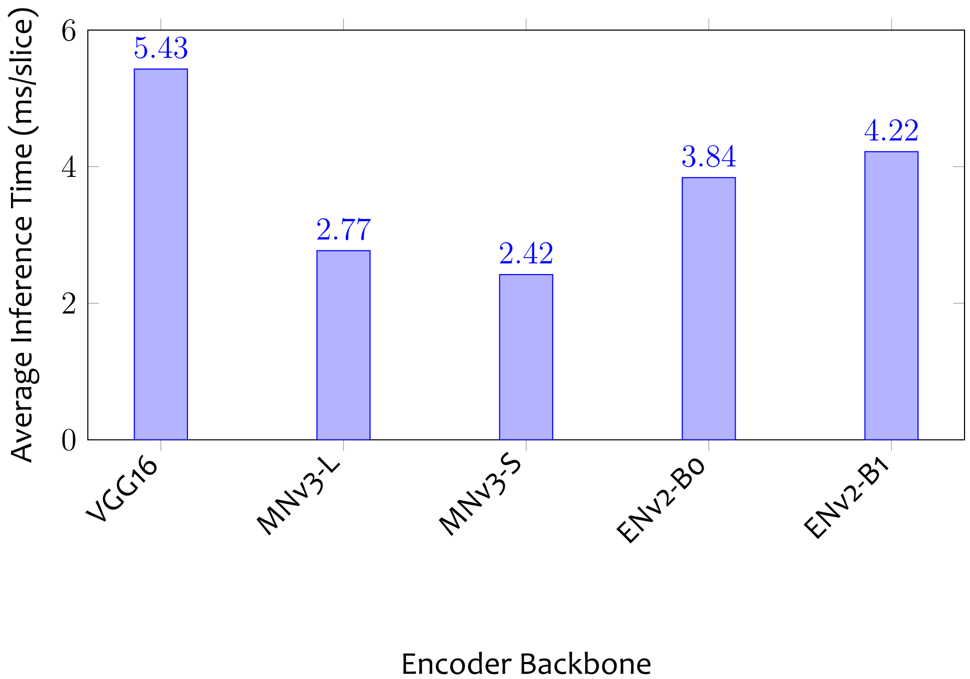

Figure 4.3: Comparison of average inference times (in milliseconds per slice) for AS-Net variants with different encoders on the test set. Lower is better.

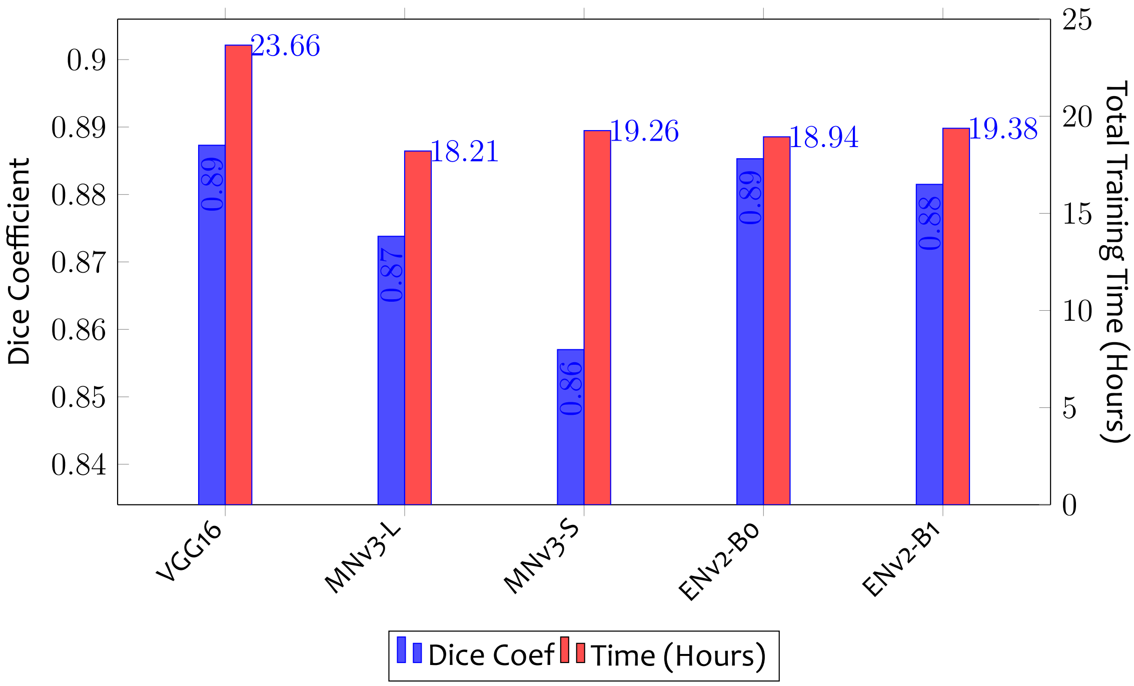

Figure 4.4: Dice Coefficient (left axis, blue) versus Total Training Time in Hours (right axis, red) for AS-Net variants.

CHAPTER 5. SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

This study investigated the optimization of the Attention Synergy Network (AS-Net) for MRI brain tumor segmentation by comparing its performance when using different encoder backbones. The original AS-Net, which employed a computationally heavy VGG16 encoder, was adapted to use lightweight alternatives: MobileNetV3 (Large and Small) and EfficientNetV2 (B0 and B1). These five variants were trained and evaluated on the BraTS 2023-GLI (Task 1) dataset, focusing on the trade-off between segmentation accuracy (Dice, IoU, etc.) and computational efficiency (total training time and average inference time per slice).

The main findings from the test set evaluations revealed that AS-Net-VGG16 yielded the highest Dice score (0.8873), establishing a strong baseline. However, AS-Net-EfficientNetV2-B0 achieved nearly identical accuracy (Dice 0.8853, only a 0.2% difference), demonstrating the competitiveness of modern lightweight architectures for this complex task. All lightweight variants significantly reduced training time compared to VGG16 (by 18-23%, equating to 4.2-5.5 hours of saved time on the V100 GPU), with MobileNetV3-Large being the fastest to train (18h 12m). Furthermore, all lightweight models showed substantial improvements in inference speed over VGG16 (5.43 ms/slice), with MobileNetV3-Small being the fastest (2.42 ms/slice, a 55% speed-up). EfficientNetV2-B0 offered the best overall balance, combining near-peak accuracy with significant efficiency gains in training (20% faster) and inference (29% faster). The study also confirmed the effectiveness of the AS-Net attention synergy mechanism (SAM, CAM, Synergy) across these diverse encoders for enhancing feature representation in MRI brain tumor segmentation.

5.2 Conclusion

Based on the empirical results obtained from evaluating the AS-Net variants on the BraTS 2023-GLI (Task 1) test set, this study draws several key conclusions regarding the optimal configuration of AS-Net for MRI brain tumor segmentation:

- **Accuracy vs. Efficiency Trade-off is Key:** The study quantifies the trade-off. While the VGG16 encoder provided marginally superior segmentation accuracy (Dice score 0.8873), modern lightweight encoders, particularly EfficientNetV2-B0, achieved highly competitive accuracy (Dice 0.8853) while offering substantial improvements in computational efficiency. This highlights that near-optimal accuracy can be achieved without the heavy computational burden of older architectures.

- **EfficientNetV2-B0 Offers Optimal Balance:** For this specific task and dataset, EfficientNetV2-B0 emerged as the most suitable lightweight encoder for AS-Net. It struck an excellent balance, delivering segmentation performance nearly identical to VGG16 while significantly reducing computational cost (training time reduced by 20% to 18h 56m, and inference time reduced by 29% to 3.84 ms/slice). This makes it a convenient choice for deployment.

- **MobileNetV3 Variants Prioritize Speed:** MobileNetV3-Large demonstrated the shortest training time (18h 12m), making it a potential option if rapid model development or retraining is the absolute priority. However, this came at the cost of a slightly larger drop in accuracy (Dice 0.8738) compared to EfficientNetV2-B0. MobileNetV3-Small achieved the fastest inference speed (2.42 ms/slice) but exhibited the lowest accuracy (Dice 0.8570), suggesting it might be too lightweight for maintaining high fidelity in this complex segmentation task.

- **AS-Net Synergy is Adaptable and Effective:** The core attention synergy mechanism (SAM, CAM, Synergy) of AS-Net proved effective when paired with various encoders, consistently contributing to strong segmentation performance across different architectural backbones. This suggests the mechanism's robustness and applicability for enhancing relevant features in MRI brain tumor segmentation, regardless of the specific encoder used.

- **Lightweight Encoders Enhance Practicality:** Replacing the heavy VGG16 encoder with efficient alternatives like EfficientNetV2 significantly enhances the practical viability of the AS-Net architecture for clinical research and potential deployment. This addresses the computational limitations noted in the original AS-Net paper (K. Hu et al., 2022), making advanced attention-based models more accessible for real-world applications.

Optimizing AS-Net with a carefully chosen lightweight encoder like EfficientNetV2-B0 allows for developing highly accurate and computationally efficient MRI brain tumor segmentation models. This represents a significant step towards bridging the gap between state-of-the-art research models and practical clinical tools, offering a compelling balance between performance and resource utilization.

5.3 Recommendations

Based on the study's findings and conclusions, the following recommendations are proposed for practical application and future research directions, aiming to further enhance the utility and performance of AS-Net for medical image segmentation:

For Practical Implementation:

- Prioritize **AS-Net-EfficientNetV2-B0** when seeking an optimal balance between high accuracy (Dice 0.8853, nearly matching VGG16) and significantly reduced computational cost (20% faster training, 29% faster inference than VGG16). This configuration offers an efficient solution for clinical research and potential deployment where accuracy and efficiency are important considerations. This aligns with findings from Tan and Le (2021) who demonstrated EfficientNetV2's superior parameter efficiency in medical imaging tasks.

- Select **AS-Net-VGG16** (Dice 0.8873) only if maximizing segmentation accuracy is the absolute priority, and the associated higher computational resources (longer training time, slower inference) are readily available and acceptable. However, as Canziani et al. (2017) noted, VGG16's high FLOPs may limit scalability in production environments.

- Consider **AS-Net-MobileNetV3-Large** (Dice 0.8738) if the fastest possible training time (18h 12m) is paramount for rapid prototyping or frequent retraining, and a slight decrease in accuracy compared to EfficientNetV2-B0 is acceptable. Its inference speed (2.77 ms/slice) is also excellent. Howard et al. (2019) showed MobileNetV3's effectiveness in resource-constrained scenarios.

- Avoid AS-Net-MobileNetV3-Small for applications demanding high accuracy due to its lower comparative performance (Dice 0.8570), despite its fastest inference speed (2.42 ms/slice). This aligns with Pavithra et al. (2023)'s findings that smaller variants may sacrifice too much accuracy for complex tasks like brain tumor segmentation.

- Validate the chosen model thoroughly in the specific clinical workflow and target hardware environment before deployment, considering the study's limitations (dataset, preprocessing, 2D approach, slice-based split). Nguyen et al. (2024) emphasize the importance of rigorous clinical validation for AI models in radiology to ensure safety and efficacy.

For Further Research:

1. **Generalizability Assessment:** CConduct extensive validation of the top-performing variants (AS-Net-VGG16 and AS-Net-EfficientNetV2-B0) on diverse, multi-center datasets (e.g., other BraTS years, different imaging protocols, patient populations) using patient-level data splits to rigorously assess robustness and confirm generalizability beyond the specific conditions of this study. Bakas et al. (2021) highlight the importance of multi-institutional validation for brain tumor segmentation models.

2. **Multi-Class Segmentation:** Adapt the AS-Net framework (output layer, loss function) to segment distinct tumor sub-regions (e.g., Necrotic and Non-Enhancing Tumor core (NCR/NET), Edema (ED), Enhancing Tumor (ET)) for enhanced clinical utility, providing more detailed information for treatment planning. Isensee et al. (2018) suggested that the hierarchical approach could be particularly informative.

3. **Benchmarking Against State-of-the-Art:** Compare the optimized AS-Net variants (especially EfficientNetV2-B0) against current state-of-the-art models like nnU-Net (Isensee et al., 2021) and recent Transformer-based architectures using standardized BraTS evaluation protocols and potentially the official BraTS online evaluation platform for a direct performance comparison. The evaluation framework proposed by Menze et al. (2014) provides a robust methodology.

4. **3D Architecture Development:** Explore implementing the AS-Net attention synergy concept within a fully 3D CNN architecture (e.g., based on 3D U-Net or V-Net) to leverage the volumetric context inherent in MRI data, potentially improving segmentation accuracy, particularly at slice boundaries. Myronenko (2018)'s work on 3D brain tumor segmentation could serve as inspiration.

5. **Encoder Exploration:** Investigate newer or larger lightweight encoders (e.g., larger EfficientNetV2 variants like B1/B2/B3, ConvNeXt (Z. Liu et al., 2022) MobileNetV4) within the AS-Net framework to explore if further improvements in the accuracy-efficiency balance are possible.

6. **Optimization and Ablation:** Perform fine-grained hyperparameter tuning tailored explicitly for each encoder (learning rate schedules, optimizer parameters, loss weights) and conduct detailed ablation studies to precisely isolate and quantify the contributions of the SAM, CAM, and Synergy modules to overall performance with different backbones. The methodology used by K. Hu et al. (2022) in their original AS-Net paper could be extended.

7. **Preprocessing Investigation:** Systematically study the impact of alternative MRI preprocessing techniques (e.g., different normalization methods like histogram matching, bias field correction, skull stripping variations) on the performance and robustness of the AS-Net models. The normalization strategies discussed by Isensee et al. (2021) could be particularly relevant.

These recommendations build upon our empirical findings and established best practices in the field. As Litjens et al. (2017) emphasized, translating deep learning models from research to clinical practice requires careful consideration of technical performance and practical constraints, including computational efficiency during training and inference. The suggested directions aim to advance AS-Net's development along both dimensions, ultimately contributing to more effective and accessible tools for brain tumor analysis.